

Addition and subtraction of algebraic terms



1

a $4x + 4y$

c $3x + 2y$

e $21xy$

g $20p^2 - 11p$

i $3x + 19y$

k $7x^2 + x$

m $25y - 9y^2$

o $3ab + 2c$

q $25x + 2y$

s $5m + 12n$

b $15m + 19n$

d $25m + 8n$

f $8q$

h $3n^2 - 4m + 2m^2$

j $18a - 8b$

l $8x - 7y + 4z$

n $11a - 7b + 4c$

p $23j + 2k$

r $12y$

t $6s - 2t$

2

a $-3x - y + 10z$

c $12x^2 + 9x + 5$

e $11mn - 2m$

g $st + 7s + 7t$

i $8x + 6y$

k $2pq + 11p^2q$

m $20xyz + 23xy$

o $-6rk + 3r - 6k$

q $-5n + 7$

b $-6xy + 2yz + 6xz$

d $-16x + 10y + 41$

f $24x^2 + 5x - 15$

h $19ab + 3bc - 9b$

j $7p + 7$

l $17x^2$

n $-12x - 4y$

p $-14vw - 2v^2w$

r $-2n + 5$

Multiplication of algebraic terms

3

a $60u$

b $-21u$

c $10x$

d $33y$

e $-72x$

f $24u$

g $-6w$

h $-30y$

4

a x^2

b $6x^2$

c $4y^2$

d $15a^2$

e $16x^2y$

f $27b^2$

g $100a^2$

h $16x^2$

5

a $72mn$

b $24wz$

c $50rs$

d $54rs$

e $42u^2v^8$

f $224p^3q^3$

g $8ab$

h $-21wz$

i $20rs$

j $-60r^8s^7$

k $-12p^5q^5$

l $-10a^4b^3$

6

a $96rs$

b $630wxy$

c $-80wxy$

d $-480hkrs$

e $-90xy$

f $-80a^3$

g $90x^2y$

h $-40hijk$



Division of algebraic terms

7

a x

b $3v$

c $\frac{m}{4}$

d $\frac{1}{4}$

e xy

f $7q$

g $\frac{4n}{5}$

h $\frac{6}{w}$

i $\frac{1}{q}$

j $10v^4$

k $-6a$

l -11

m $-\frac{1}{11}$

n -4

o $\frac{4}{9}$

p $-ac$

q $-\frac{1}{j}$

r $-\frac{6}{k}$

s $-\frac{1}{b}$

t $-\frac{2}{3}$

u $-3w^5$

v 12

8

a $\frac{m}{8}$

b 5

c 2

d -4

e $-11s$

f 6

g $2n$

h $3x$

i $3r$

j -2

k $11x$

l $10c$

9

a 6

b 2

c $3x^2$

d 14

e $3a$

f $\frac{9}{2}$

g $18x$

h $-4x$

i $33xy$

j $15y^2$

k $10x$

l y

Applications

10 $(10 + a)$ cm

11

a 24



b $6y$

12

a $3x$ cm

b The water level is increasing by the same amount each hour.

13 $5p + 10q$

14

a $(300 + 70t)$ m

b Uther is paddling at a constant pace over the first t minutes.

15

a $9 + y$

b It represents that it takes Sally 9 minutes longer than the bus trip to get to school.

16

a $44WL$ cm³

b $132WLH$ cm³

c $3H$

d 264

17

a $6xy$ cm³

b $48xyz$ cm³

c $8z$

d 120